

MOHSEN FEKRI

AI & DATA | MACHINE LEARNING | COMPUTER VISION

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PROFESSIONAL SUMMARY

AI and machine learning practitioner who builds end-to-end: faculty-led research in spatio-temporal medical-image segmentation, credit-risk models designed for human review, and enterprise RAG with page-level citations and role-based access. Comfortable where models meet production — imbalanced data, evaluation trade-offs, model drift, and inference optimization. Pursuing a BAsC in Computer Science for Software Development (Co-op) at York University, Lassonde School of Engineering. Aligned to EY AI & Data, Digital Engineering, and Technology Risk, with working knowledge of EY.ai, EYQ, FlexiGenAI, and Responsible AI.

TECHNICAL SKILLS

AI & Machine Learning: PyTorch, MONAI, computer vision, segmentation, classification, spatio-temporal modelling, model evaluation (precision, recall, F1, ROC-AUC), imbalanced data, model drift, explainable AI

GenAI & Agents: RAG, MCP (Model Context Protocol), OpenAI-compatible API standard, vector databases, LangChain/LlamaIndex, RAGAS evaluation, Responsible AI

Engineering & Data: Python, SQL, Java, JavaScript, Pandas (DataFrames), NumPy, Scikit-learn, XGBoost, FastAPI, Streamlit, Gradio, Docker, Kubernetes, TensorRT, Git

AI RESEARCH EXPERIENCE

Undergraduate Research Award (UGRA) & Teaching Assistant | York University | 2025

Supervisor: Prof. Navid Mohaghegh — Spatio-Temporal Ultrasound Needle Segmentation

- Held a **UGRA** research award for AI-assisted needle localization in ultrasound, alongside teaching assistant duties for Prof. Mohaghegh.
- Built a PyTorch and MONAI segmentation pipeline pairing Swin Transformer **spatial** feature extraction with ConvLSTM **temporal** modelling across frames.
- Drove preprocessing, model development, and frame-level evaluation on sequential ultrasound, scoring segmentation quality with Dice and IoU.

SELECTED AI PROJECTS

AI Credit Risk & Loan Decision Support System | Python | Scikit-learn | XGBoost | Explainable AI | 2026

- Modelled borrower credit history and financial attributes to estimate loan eligibility and default risk.
- Tuned against a heavily **imbalanced** default population, weighing false positives against false negatives across precision, recall, F1, ROC-AUC, and confusion-matrix analysis.
- Surfaced the features driving each decision and framed output as decision support, keeping the final lending call with a human reviewer.

Enterprise Document Question-Answering Assistant | Python | RAG | MCP | FastAPI | Gradio | RAGAS | 2026

- Shipped a RAG assistant over company policies, audit procedures, risk reports, and cybersecurity documentation in PDF and Word.
- Exposed retrieval as **MCP** tools behind an OpenAI-compatible API so agentic clients can drive the assistant directly.
- Returned page-level citations and abstained when the corpus lacked evidence, rather than answering from thin retrieval.
- Tuned chunking and retrieval against **RAGAS** faithfulness scores, behind role-based permissions.

Medical Image Cancer Detection Prototype | Python | PyTorch | EfficientNet-B0 | TensorRT | Streamlit | 2025

- Trained a CNN classifier that standardizes images to 224 x 224 RGB, with a sigmoid probability and an adjustable decision threshold.
- Cut model size and inference latency with **TensorRT** quantization and pruning, tracking the accuracy trade-off at each step.
- Added visual interpretability for reviewer oversight, labelled clearly as educational and research-only rather than diagnostic.

EDUCATION

York University, Lassonde School of Engineering | Toronto (Markham Campus) | Expected 2029

Bachelor of Applied Science (BASc), Computer Science for Software Development, Co-op | GPA: 6.7/9.0

ADDITIONAL EXPERIENCE

Operations Assistant | York University | **May 2026 - Sep 2026**

Program & Administrative Assistant | YMCA of Greater Toronto | **Dec 2024 - Jan 2026**

IT Support & Lab Assistant, Volunteer | Toronto Public Library | **Mar 2023 - Sep 2023**

CERTIFICATION IN PROGRESS

AI-103: Developing AI Apps and Agents on Azure | Microsoft | **In Progress**